

```
self.x = random.randrange(1, maxWidth - 100)
self.y = random.randrange(1, maxHeight - 100)
```

Finally, each class sets the instance variable `self.shapeType` to an appropriate string.

Whenever we find a set of classes that implement the exact same method and/or share some code in a commonly named method, we should recognize that this is a good candidate for inheritance.

Let's extract the common code from the three classes and build a common base class named `Shape`, shown in [Listing 10-6](#).

File: InheritedShapes/ShapeBasic.py

```
# Shape class - basic

import random

# Set up the colors
RED = (255, 0, 0)
GREEN = (0, 255, 0)
BLUE = (0, 0, 255)

class Shape():

    ❶ def __init__(self, window, shapeType, maxWidth, maxHeight):
        self.window = window
        self.shapeType = shapeType
        self.color = random.choice((RED, GREEN, BLUE))
        self.x = random.randrange(1, maxWidth - 100)
        self.y = random.randrange(25, maxHeight - 100)

    ❷ def getType(self):
        return self.shapeType
```

Listing 10-6: The `Shape` class, to be used as a base class

The class consists of only two methods: `__init__()` and `getType()`. The `__init__()` method ❶ remembers the data passed in in instance variables, then randomly chooses a color and a starting location (`self.x` and `self.y`).